A Dictionary Learning Approach for Multi-Class Abnormality Classification and Grounded Localization in 3D Chest CT

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Abstract

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Acronyms

AC	-	Alternating Current
DC	-	Direct Current
LV	-	Low Voltage
MV	-	Medium Voltage

Chapter 1

Introduction

Developing a unified architectural pipeline for multi-class abnormality classification and grounded localization in 3D Computed Tomography (CT)a ddresses two of the most critical imperatives in clinical AI adoption: diagnostic accuracy and explainability. Specifically, utilizing dense pixel-level segmentation as the mechanism for localization forces the architecture to provide voxel-level visual evidence for its diagnostic claims. This ensures that the model is held accountable and its reasoning aligns with clinical diagnosis.

1.1 Background

The transition from 2D projectional radiography to volumetric CT represents a paradigm shift in medical image analysis. For Lung disease diagnosis, Chest X-rays (CXR) are commonly used as the first imaging test due to its low cost and low radiation exposure. However, studies show that their capability to detect abnormailities is limited when compared to 3D CT. [**?**].

CXR has historically benefited from the availability of massive, radiologist-annotated datasets, while the analysis of 3D chest CT volumes has been severely constrained by a lack of such data. However, the recent introduction of large-scale, text-paired datasets has facilitated the development of state-of-the-art (SotA) vision-language foundation models tailored specifically for volumetric imaging. [**?**] [**?**] [**?**].

Beyond deep learning approaches, the Label Consistent K-SVD (LC-KSVD) algorithm offers a viable approach for interpretable CT analysis. LC-KSVD is a dictionary learning algorithm that incorporates label information directly into the dictionary construction process [**?**], making it inherently discriminative and well-suited for multi-abnormality tasks.

1.2 Problem Statement

The clinical utility of Deep Learning-based approaches is hindered by a lack of interpretability. Tools like Grad-CAM provide visual heatmaps highlighting important regions in an input image by analyzing gradients from the final convolutional layer [**?**]. These post-hoc explanations are often insufficient for medical professionals who require a granular understanding of the model's decision making process. In high stakes environments, a model that produces a diagnosis without transparency in its reasoning is difficult to integrate into standard workflows, as it lacks the accountability necessary for legal and ethical compliance.

1.3 Objectives and Scope

The primary goal of this project is to address the limitations of existing Deep Learning based approaches for Lung abnormality classification and localization by developing an inheerntly interpretable dictionary learning framework based on LC-KSVD. This approach aims to enable transparent, anatomically grounded decision making, where each classification outcome can be traced back to specific pathological patterns. This aligns with clinical reasoning and facilitates validation and adoption in radiological workflows. The specific objectives are as follows:

1.3.1 Objectives

- Implement a Dictionary Learning approach to learn normal and pathological dictionary atoms using a publicly available annotated 3D chest CT dataset.
- Map dictionary atoms back to anatomical features and visualize their spatial contribution maps.
- Benchmark model performance against state-of-the-art classification and segmentation models on the same dataset.
- Evaluate abnormality localization by the model on a real-world dataset with expert radiologist review.

1.3.2 Scope

The scope of this study focuses on the development and evaluation of a dictionary learning based approach for classification and grounded localization of lung abnormalities using volumetric 3D CT data, with the following limitations:

- This study is limited to 3 thoracic abnormalities: pulmonary nodules, ground-glass opacities, and consolidations.
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1.4 Literature Review

The landscape of 3D CT analysis is currently divided into three primary architectural paradigms:

- Classification-based models that prioritize diagnostic labels
- Segmentation-based models that prioritize boundary precision
- Unified architectures designed for grounded localization

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SotA models are predominantly specialized in either classification or segmentation. However, recent systems, particularly in 2025 and 2026, are moving toward multi-task learning or foundation models that can perform both diagnosis and voxel-level segmentation simultaneously.

The efficacy of these models is increasingly evaluated on large-scale, multi-institutional benchmarks such as CT-RATE, FLARE 2024-2025 and MSWAL, which provide the complexity necessary to test generalizability across diverse patient populations and scanner manufacturers.

1.4.1 Classification Architectures

Classification-based models for 3D CT have shifted away from 3D Convolutional Neural Networks (CNNs) such as 3D U-Net, 3D ResNet and DenseNet pipelines, toward Vision Transformers (ViTs) and self-supervised encoders. These models are designed to identify the presence of one or more abnormalities within a full volumetric scan. The benchmark for this paradigm is the CT-RATE dataset, which introduced the first public large-scale 3D medical imaging dataset with paired textual reports, enabling the development of Contrastive Language-Image Pretraining for Computed Tomography (CT-CLIP).

(About CT-CLIP) Recent benchmarks such as Merlin and DINOv3 have further challenged the necessity of domain-specific pretraining.

Merlin is a vision-language foundation model exclusively pre-trained on abdominal CT scans. Despite it's training, Merlin has consistently demonstrated exceptional zero-shot performance on chest CT classification tasks, even when it was restricted to a linear probe, significantly outperforming models like CT-CLIP, M3FM, and CT-FM that were explicitly trained on thoracic data, by an average of 12.3% in Area Under the Receiver Operating Characteristic Curve (AUROC). Specifically, in cross-domain retrieval tasks, Merlin surpassed CT-CLIP by 24.7%. DINOv3, a self-supervised ViT pretrained on over 1.7 billion natural images, outperform specialized medical models like CT-Net and CT-CLIP across all metrics on several classification tasks due to it's ability to capture fine-grained textural features.

1.4.2 Segmentation Architectures

The prevailing SotA for supervised volumetric segmentation remains variations of the 3D U-Net, such as nnU-Net, which utilizes an adaptive framework to optimize preprocessing and architecture for specific datasets. However, the field has recently shifted toward foundation models capable of universal segmentation such as SegVol and SAM-Med3D. The SegVol model is the first 3D foundation model specifically designed for universal volumetric medical image segmentation. Similarly, SAM-Med3D adapts the Segment Anything Model (SAM) for 3D medical data.

Recent evaluations published at the Machine Learning for Healthcare (MLHC) Conference in 2025, which specifically utilized the ReXGroundingCT dataset, revealed that the current SotA text-prompted segmentation models that demonstrate near-perfect performance in general computer vision tasks struggle with the nuanced clinical descriptions and ill-defined boundaries of chest CT abnormalities.

1.4.3 Unified Architectures

Recent models have been designed to integrate abnormality detection, segmentation, and multimodal chain-of-thought reasoning, enabling pixel-level localization, structured report generation, and physician-like diagnostic inference in a single framework. BiomedParse, and it's production-ready, version MedImageParse developed by Microsoft, are foundation models designed to parse 3D medical images. Other notable models include Citrus-V, MTMed3D and MedVista3D, where each architecture incorporates a multi-task learning (MTL) framework.

1.4.4 Dictionary Learning Approaches

Applications of sparse dictionary learning in medical imaging have demonstrated significant success across diverse diagnostic tasks. In histopathological analysis, Discriminative Feature oriented Dictionary Learning (DFDL) has shown the ability to learn class specific dictionaries that achieve high classification accu- racy even with limited training data a critical advantage when generous training samples are not available[19]. In neuroimaging, multi feature kernel supervised dictionary learning approaches have demonstrated superior performance in distinguishing Alzheimer's disease patients from cognitively unimpaired individuals, achieving 98.18% classification accuracy while maintaining fast testing times[20].

Most relevant to our work, dictionary learning has been successfully applied to lung disease diagnosis from CT images. One approach utilized both frozen dic- tionary learning and label consistent LC-KSVD to classify COVID 19 CT scans, achieving 89% accuracy on the CC-CCII dataset[21]. The framework's key ad- vantage lies in its interpretability, where the sparse coding output can be directly mapped back to the original input signal domain. This work demonstrates that sparse representation methods offer an effective balance between performance and explainability for lung disease classification tasks, providing clear feature represen- tations while maintaining competitive accuracy. Similarly, sparse representation methods have achieved 95.4% accuracy in classifying diffuse lung disease pat- terns on HRCT images, effectively handling the high variety and complexity of patterns[22].

1.4.5 Explainability Approaches

Explainability methods are essential in chest CT analysis because abnormality classification alone is not sufficient for clinical adoption [?]. For this project, the most relevant explainability techniques are those that can relate predictions back to anatomical regions and, ideally, support voxel-level interpretation in volumetric data.

Grad-CAM is one of the most widely used post-hoc explanation methods for CNN-based classifiers [?]. It generates class-discriminative heatmaps by using gradients flowing into the final convolutional layer, allowing the user to identify image regions that influenced a prediction [?]. In medical imaging, Grad-CAM has been applied to chest radiographs and CT classification tasks to provide coarse visual explanations [?]. However, its main limitation is that it produces low-resolution saliency maps and does not directly provide precise lesion boundaries or voxel-level localization, which is a major drawback for 3D CT interpretation [? ?].

Seg-Grad-CAM extends this idea by adapting Grad-CAM to semantic segmentation outputs [?]. Instead of only highlighting regions that are important for image-level classification, it produces pixel-level relevance heatmaps for segmentation predictions [?]. This is particularly useful for volumetric medical imaging tasks where the target abnormality occupies a sparse region within a large 3D scan. For a project focused on grounded localization, Seg-Grad-CAM is more relevant than standard Grad-CAM because it better reflects whether the model is attending to the pathological structure rather than surrounding tissue.

HiRes-CAM addresses one of the key weaknesses of Grad-CAM by producing higher-resolution and more faithful localization maps [?]. This is important in chest CT because many abnormalities, such as small nodules, consolidations, or subtle interstitial changes, may be missed by coarse heatmaps. HiRes-CAM aims to preserve fine spatial detail, making it more suitable for medical imaging applications where precise localization is required [?]. Nevertheless, like other gradient-based methods, it remains a post-hoc explanation technique and does not change the internal decision process of the model.

In the context of this project, these methods are useful baselines for evaluating interpretability in deep learning models. However, they remain fundamentally different from dictionary learning approaches such as LC-KSVD, which provide intrinsic interpretability by representing predictions through sparse combinations of learned atoms [? ?]. Compared with Grad-CAM, Seg-Grad-CAM, and HiRes-CAM, LC-KSVD offers a more direct and anatomically traceable explanation mechanism for multi-abnormality classification and localization in 3D CT.

1.5 Report Outline

Chapter 2

Background

2.1 LC-KSVD

The LC-KSVD algorithm solves an optimization problem that balances reconstruction accuracy with label consistency and classification error. Given a set of image patches P, the goal is to learn a dictionary D and sparse coefficients X that minimize the following objective:

$$\min_{D,X,A,W} \|P - DX\|_F^2 + \alpha\|Q - AX\|_F^2 + \beta\|H - WX\|_F^2 \quad \text{s.t.} \quad \forall i, \|x_i\|_0 \leq T$$

The first term $\|P - DX\|_F^2$ is the standard K-SVD reconstruction error. The second term $\|Q - AX\|_F^2$ introduces the "discriminative sparse-code error," where Q is a label-consistent constraint matrix that encourages patches from the same class to activate the same dictionary atoms. The third term $\|H - WX\|_F^2$ is the classification error, where W is a linear classifier matrix.

This formulation ensures that the dictionary atoms are not just representative of the image data but are also class-specific. In 3D CT multi-abnormality segmentation, this allows the model to learn distinct "sub-dictionaries" for each abnormality, facilitating easier classification of individual voxels.

2.2 Sparse Representation, K-SVD, and Frozen Dictionary Learning

Sparse representation is a signal modeling technique where a CT image patch is expressed as a linear combination of a small number of basis elements called dictionary atoms. Instead of using all available features, the signal is approximated as:

$$y = Dx$$

where D is the dictionary and x is a sparse coefficient vector with mostly zero values. This leads to a compact and interpretable representation.

The dictionary D is typically learned using K-SVD (K-Singular Value Decomposition), which solves the following optimization problem:

$$\min_{D,X} \|P - DX\|_F^2 \quad \text{s.t.} \quad \forall i, \|x_i\|_0 \leq T$$

where P represents the training data, X is the sparse coefficient matrix, and T controls the sparsity level. K-SVD alternates between sparse coding and dictionary update steps to minimize reconstruction error. However, it does not incorporate class labels, which limits its discriminative ability for classification tasks.

To address this limitation, LC-KSVD extends the standard K-SVD framework by incorporating class label information into the dictionary learning process. It jointly learns a dictionary, sparse codes, and a classifier so that the learned representations are not only good for reconstruction but also discriminative for classification tasks through label consistency constraints. However, LC-KSVD still learns a single shared dictionary for all classes.

To overcome this limitation, Frozen Dictionary Learning introduces a two-stage training strategy. In the first stage, a dictionary is learned from normal CT scans to capture healthy anatomical structures. In the second stage, this dictionary is frozen and kept fixed, while additional atoms are learned from abnormal scans only. This formulation can be interpreted as:

- preserving D_{normal} learned from healthy data - learning D_{abnormal} independently for disease patterns

This separation ensures that normal anatomical structures are not overwritten, while abnormal features are explicitly isolated, improving interpretability and stability.

2.3 Radiological Patterns in Lung CT Imaging

2.3.1 CT Imaging

Computed Tomography (CT) imaging is widely used in lung disease diagnosis due to its ability to generate detailed cross-sectional images of lung structures. CT scans provide high-resolution visualization of abnormalities such as Ground Glass Opacity (GGO), consolidation, and pulmonary nodules, enabling accurate assessment and early detection of lung diseases.

In modern medical imaging systems, CT imaging also supports computer-aided diagnosis and deep learning-based analysis for automated detection, segmentation, and classification of lung abnormalities. These capabilities improve diagnostic efficiency and assist radiologists in clinical decision-making.

2.3.2 Ground Glass Opacity (GGO) and Consolidation

Ground Glass Opacity (GGO) and consolidation are two important radiological patterns used in lung CT diagnosis.

Ground Glass Opacity (GGO):

- Appears as hazy regions in CT images
- Does not fully obscure underlying blood vessels or airway structures
- Typically indicates partial filling of air spaces, inflammation, or early-stage infection

Consolidation:

- Appears denser than GGO
- Completely obscures underlying lung structures such as vessels and airway walls
- Often associated with severe infection, fluid accumulation, or advanced disease stages

The transition between GGO and consolidation is clinically important, as it can indicate disease progression or worsening lung conditions.

2.3.3 Pulmonary Nodules

Pulmonary nodules are small, rounded lesions commonly observed in lung CT scans and are important indicators for early disease detection.

Characteristics of Nodules:

- Small, well-defined or slightly irregular rounded opacities
- Usually less than 3 cm in diameter
- Can appear solid, subsolid, or ground-glass in nature

Clinical Significance:

- May represent benign conditions such as infections or scars
- Can also indicate early-stage lung cancer in suspicious cases
- Requires careful monitoring of size, shape, and growth over time

Pulmonary nodules are critical in early screening and risk assessment, especially in automated CT analysis systems.

2.4 Explainable Artificial Intelligence (XAI) in Medical Imaging

Explainable AI (XAI) in medical imaging is used to make deep learning predictions more interpretable, especially in clinical applications where understanding model decisions is important.

Grad-CAM is a widely used method that highlights important regions in an image by generating a heatmap based on the gradients flowing into the final convolutional layers. In CT imaging, it helps identify which lung regions contributed most to a prediction, such as areas with abnormalities or infections.

HiResCAM is an improved version of activation-based visualization methods. It produces more detailed and higher-resolution heatmaps compared to Grad-CAM, allowing better localization of small or subtle abnormalities in medical images.

Both methods are post-hoc explanation techniques, meaning they are applied after model training. They do not modify the model itself but instead help interpret its predictions by visualizing the regions of the input image that most influenced the output.

2.5 Vision Transformers (ViTs) in Medical Imaging

Vision Transformers (ViTs) are deep learning architectures that apply the transformer mechanism, originally developed for Natural Language Processing (NLP), to image analysis tasks, Vision Transformers divide an image into smaller patches and process them as sequential tokens.

Given an input image I, the image is divided into fixed-size patches. Each patch is flattened and projected into an embedding vector. Positional embeddings are then added to preserve spatial information before passing the sequence into transformer encoder layers.

The self-attention mechanism used in ViTs allows the model to capture long-range dependencies and global contextual relationships between different regions of an image. The attention operation can be represented as:

$$\mathrm{Attention}(Q, K, V) = \mathrm{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

where Q represents queries, K represents keys, V represents values, and d_k is the dimension of the key vectors.

In medical imaging, Vision Transformers have shown strong performance in tasks such as classification, segmentation, abnormality detection, and disease localization. Their ability to model global image context is particularly useful for analyzing complex CT volumes where abnormalities may span multiple regions.

For lung CT analysis, ViTs can learn meaningful representations of radiological patterns such as Ground Glass Opacity (GGO), consolidation, and pulmonary nodules. Hybrid architectures combining CNNs and Vision Transformers are also commonly used to capture both local texture information and global contextual features.

Recent developments such as Swin Transformers and 3D Vision Transformers further extend this capability by enabling hierarchical feature extraction and volumetric analysis for 3D medical imaging applications.

Chapter 3

Methodology

3.1 Data Acquisition and Preprocessing

Two public chest CT datasets were used in this work: CT-RATE and ReXGroundingCT. CT-RATE was used as the primary source of volume-level labels (normal/abnormal and abnormality presence) derived from the dataset provided multi finding annotations, while ReXGroundingCT was used to obtain lesion level spatial supervision through radiologist linked segmentation masks together with finding descriptions. In practice, CT-RATE provided the global abnormality labels needed to build classification splits, and ReXGroundingCT provided grounded masks and metadata required for localisation-oriented experiments.

After an initial review of the 18 thoracic findings available in CT-RATE, a clinical discussion with the supervising radiologist produced a shortlist of abnormalities considered both clinically meaningful and suitable for interpretable analysis (Atelectasis, Lung nodule, Ground-Glass Opacity (GGO), Consolidation, Mosaic attenuation, Bronchiectasis, and Calcification). However, after checking practical label and mask availability across the two datasets, the final experiments were restricted to three abnormalities with the most consistent coverage: *Lung nodule*, *Lung opacity / GGO*, and *Consolidation*. The remaining candidates were excluded mainly due to limited grounded lesion samples, inconsistent annotation coverage, or insufficiently distinct mask supervision for reliable training and evaluation.

The preprocessing pipeline followed the LC-KSVD implementation. CT volumes and corresponding masks were loaded from NIfTI files, then resampled to an isotropic spacing of 1.5 mm to reduce inter-scan resolution variability. Volumes were resampled with trilinear interpolation and segmentation masks with nearest-neighbour interpolation to preserve discrete labels. Intensities were windowed using a lung HU range of $[-1000, 200]$ and linearly normalised to $[0, 1]$. For dictionary learning, each preprocessed volume was converted into fixed-size 3D patches of $32 \times 32 \times 32$ voxel. Positive patches were sampled from lesion regions using the grounded masks (minimum overlap ratio 5%, with a fallback for very small lesions). Negative patches were sampled from non overlapping regions (including normal scans), and patch vectors were L_2-normalised prior to sparse coding.

3.2 Implementation of the LC-KSVD Approach

To implement the Dictionary Learning approach for our study, we required a concrete implementation of the LC-KSVD algorithm. Currently, many of the available standard implementations of KSVD and LC-KSVD are developed using MATLAB, which is not ideal for integration with our Python-based data processing and evaluation pipeline. Although A few open-source Python implementations LC-KSVD are available, they either lacked an importable library, or did not provide the flexibility required for our study.

Therefore, we developed a custom implementation of the LC-KSVD algorithm in Python and published it on PyPI under the name "reppi." The implementation follows a modular design, consisting of three separate components:

1. SRC - Utility Functions shared across Sparse Coding Algorithms
2. OMP - Implementation of Batch-OMP and OMP-Cholesky approaches for Sparse Coding
3. KSVD - Implementation of KSVD Dictionary Learning Algorithm
4. LC-KSVD - Implementation of LC-KSVD1 and LC-KSVD2 Dictionary Learning Algorithms

This design allows for flexibility in integrating alternative sparse coding methods. The source code is available at github.com/ckekula/reppi.

This modular architecture enables straightforward integration of the algorithm through library imports, while providing the flexibility required to experiment with different hyperparameters and configurations. The implementations of the SRC, OMP, and LC-KSVD modules are provided in Appendix A, Appendix B, and Appendix C, respectively.

3.3 Abnormality Classification

Abnormality classification was implemented as a set of one-vs-normal binary LC-KSVD2 models, trained separately for each target finding (Lung nodule, Lung opacity, and Consolidation). After preprocessing, each CT volume is represented as a matrix of vectorised $32 \times 32 \times 32$ patches, with each patch column-wise normalised to unit L_2 norm before sparse coding. At inference time, patch-wise sparse codes Γ are computed using OMP with the learned dictionary D, and classification is performed using the learned linear classifier weights W from LC-KSVD2 by scoring $s = W_{\text{pos}}\Gamma$. For volume-level prediction, the implementation aggregates evidence by averaging the sparse codes across all sampled patches and applies a sign threshold ($s > 0$) to obtain the final abnormality present/absent decision.

3.4 Back-Projecting Class-Discriminative Atoms

The unique intuition we propose in our approach is to generate spatial contribution maps through back-projection. Because each atom in the dictionary D is associated with a specific class label through the matrix Q, the model can theoretically "explain" its decision for a test patch by examining which atoms were used in its sparse representation.

To generate a spatial contribution map for a class c, the algorithm identifies all atoms d_k that have been assigned to that class. For a query voxel or patch with sparse coefficients x, the contribution C_c is calculated as the partial reconstruction:

$$C_c = \sum_{k \in \text{Class } c} d_k x_k$$

This process should theoretically back-project the class-specific features into the original voxel space, creating a map that highlights the regions contributing to the detection of a specific abnormality. for class-discriminative atoms operating on raw voxel-space inputs, the back-projection is an exact, closed-form spatial attribution.

3.5 Deriving Explainability through Segmentation Masks

To evaluate the clinical accuracy of LC-KSVD, these contribution maps are converted into binary segmentation masks for comparison with ground truth annotations. This conversion can be achieved through various thresholding methods.

Grad-CAM's approximation has two sources of imprecision — the gradient itself is a local linear approximation of a non-linear function, and the spatial upsampling from the coarse feature map to input resolution introduces further interpolation error. LC-KSVD's contribution map has only one source of imprecision: the thresholding step you apply to binarize it into a mask. The map itself — before thresholding — is exact given the sparse code and the dictionary.

Unlike the passive Grad-CAM approach, atom back-projection is generative. The atoms are what the model actually used to reconstruct and classify the signal. There's no proxy involved. The localization evidence is produced by the same computation that produced the classification, not by a separate post-hoc analysis run afterwards.

Chapter 4

Experimentation and Evaluation

4.1 Project Proposal

4.2 Project Progress

4.3 End Exam

Chapter 5

Discussion

Appendix A

Appendix 1

The Appendices are numbered as A, B,.. and so on. Sections and Subsections of the appendix then would be A.1, A.1.1, A.2 etc.

Appendix B

Appendix 2